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Impact of an AI prognostic tool on clinician performance in colorectal liver metastases

Qichen Chen^{1,5}✉, Jinliang Tong^{2,5}, Yiqiao Deng^{2,5}, Xinyu Bi², Yuan Li³✉, Kan Li⁴✉ & Hong Zhao²✉

While thousands of AI prediction models are published annually, few are adopted into routine practice, partly because improved statistical performance does not necessarily translate into meaningful impact on clinical decision-making. We conducted a prospective randomized multi-reader multi-case study to evaluate how a machine learning-based prognostic tool influences clinician performance in colorectal liver metastases (CRLM). In a prospective, randomized multi-reader multi-case trial (NCT07027605; Registration Date: January 1, 2025), 12 surgical oncologists assessed 166 retrospective CRLM cases with and without tool assistance in a crossed design with a 5-week washout. The primary endpoint was the difference in AUC for predicting 3-year mortality. Between January and July 2025, 12 readers completed 3984 assessments. Model assistance significantly improved the AUC for 3-year mortality prediction (mean difference 0.091; 95% CI 0.001–0.181; $P = 0.048$) and consistently improved accuracy across secondary prognostic endpoints. It also reduced decision time (2.53 vs. 3.04 minutes) and increased reader confidence. Benefits were greatest for junior to mid-level surgical oncologists. This exploratory study demonstrates that a machine learning prognostic tool can significantly improve accuracy, efficiency, and confidence in CRLM evaluation.

The rapid advancement of artificial intelligence (AI) and machine learning (ML) has led to a proliferation of clinical prediction models across medical specialties. Despite promising statistical performance, the vast majority of these models fail to translate into routine clinical practice, largely because improvements in discrimination or calibration alone do not demonstrate meaningful impact on clinical decision-making or patient outcomes^{1–9}. This implementation gap has prompted increasing attention from regulatory bodies, which now emphasize not only analytical validity but also demonstration of clinical benefit and real-world integration for AI/ML-enabled medical products^{10,11}.

Colorectal liver metastases (CRLM) represent a major source of morbidity and mortality among patients with colorectal cancer^{12,13}. For CRLM, local treatment remains the cornerstone of curative-intent management^{14,15}. However, postoperative outcomes are highly heterogeneous, highlighting the need for robust prognostic tools to support risk

stratification and individualized treatment planning^{16,17}. Although numerous prognostic models for CRLM have been developed^{18,19}, their integration into routine surgical practice has been limited, reflecting the broader translational challenges facing AI-driven prediction tools in the surgical management of CRLM.

To address this gap, we conducted a prospective, randomized multi-reader multi-case (MRMC) study (ClinicalTrials.gov NCT07027605) to evaluate whether a high-performing personalized ML-based prognostic tool for CRLM^{14,15} improves clinician performance in a clinically contextualized decision-making setting. Specifically, we sought to determine whether AI assistance enhances prognostic accuracy, decision consistency, efficiency, and confidence among surgical oncologists. To our knowledge, this is the first exploratory study to systematically assess the impact of an AI-based prognostic model on clinician performance in the surgical management of CRLM.

¹Department of Colorectal Surgery, National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China. ²Department of Hepatobiliary Surgery, National Cancer Center/National Clinical Research Center for Cancer/Cancer Hospital, Chinese Academy of Medical Sciences and Peking Union Medical College, Beijing, China. ³Department of Colorectal Surgery, State Key Laboratory of Oncology in South China, Guangdong Provincial Clinical Research Center for Cancer, Sun Yat-sen University Cancer Center, Guangzhou, China. ⁴Daiichi Sankyo, NJ, Basking Ridge, NJ, USA. ⁵These authors contributed equally: Qichen Chen, Jinliang Tong, Yiqiao Deng. ✉e-mail: chenqichen0822@126.com; liyuan@sysucc.org.cn; kan.li@daichisankyo.com; zhaohong@cicams.ac.cn

Table 1 | Assessment distributions under the aided condition vs. the unaided condition

Variable	Assessment Categories	Aided	Unaided	Nominal <i>P</i> -value*
1-year recurrence risk, <i>n</i> (%)	low risk, less than 10%	755 (37.9%)	736 (36.9%)	<0.001
	moderate risk 10–30%	109 (5.5%)	800 (40.2%)	
	high risk 30–70%	745 (37.4%)	450 (22.6%)	
	critical risk greater than or equal to 70%	383 (19.2%)	6 (0.3%)	
3-year recurrence risk, <i>n</i> (%)	low risk, less than 10%	373 (18.7%)	67 (3.4%)	<0.001
	moderate risk 10–30%	94 (4.7%)	650 (32.6%)	
	high risk 30–70%	176 (8.8%)	1051 (52.8%)	
	critical risk greater than or equal to 70%	1349 (67.7%)	224 (11.2%)	
5-year recurrence risk, <i>n</i> (%)	low risk, less than 10%	125 (6.3%)	11 (0.6%)	<0.001
	moderate risk 10–30%	288 (14.5%)	196 (9.8%)	
	high risk 30–70%	48 (2.4%)	496 (24.9%)	
	critical risk greater than or equal to 70%	1531 (76.9%)	1289 (64.7%)	
1-year mortality risk, <i>n</i> (%)	low risk, less than 10%	1635 (82.1%)	1287 (64.6%)	<0.001
	moderate risk 10–30%	113 (5.7%)	542 (27.2%)	
	high risk 30–70%	174 (8.7%)	163 (8.2%)	
	critical risk greater than or equal to 70%	70 (3.5%)	0 (0%)	
3-year mortality risk, <i>n</i> (%)	low risk, less than 10%	760 (38.2%)	299 (15.0%)	<0.001
	moderate risk 10–30%	154 (7.7%)	937 (47.0%)	
	high risk 30–70%	404 (20.3%)	602 (30.2%)	
	critical risk greater than or equal to 70%	674 (33.8%)	154 (7.7%)	
5-year mortality risk, <i>n</i> (%)	low risk, less than 10%	358 (18.0%)	5 (0.3%)	<0.001
	moderate risk 10–30%	390 (19.6%)	510 (25.6%)	
	high risk 30–70%	78 (3.9%)	659 (33.1%)	
	critical risk greater than or equal to 70%	1166 (58.5%)	818 (41.1%)	
Support surgery, <i>n</i> (%)	Yes	1779 (89.3%)	1813 (91.0%)	0.009
	No	213 (10.7%)	179 (9.0%)	
Post-operative follow-up Timing recommendation, <i>n</i> (%)	Post-operative 1 month	1207 (60.6%)	1191 (59.8%)	<0.001
	Post-operative 2 months	256 (12.9%)	503 (25.3%)	
	post-operative 3 months	509 (25.6%)	276 (13.9%)	
	Post-operative 4 months	0 (0%)	0 (0%)	
	Post-operative 5 months	0 (0%)	1 (0.1%)	
	Post-operative 6 months	20 (1.0%)	21 (1.1%)	
Confidence levels, <i>n</i> (%)	Low confidence	0 (0%)	164 (8.2%)	<0.001
	Moderate confidence	51 (2.6%)	725 (36.4%)	
	High confidence	895 (44.9%)	971 (48.7%)	
	Extremely high confidence	1046 (52.5%)	132 (6.6%)	
Completion time, mean(SD), min	–	2.53 (0.632)	3.04 (0.679)	<0.001

*Statistical Comparisons were performed using the Wilcoxon signed-rank test for the continuous variable of completion time and the McNemar-Bowker test for all other paired categorical assessments.

Results

Case characteristics

Between January 2025 and July 2025, 12 surgical oncologists (readers), stratified by experience level (4 junior, 4 mid-level, and 4 senior), evaluated 166 retrospective patients (cases) with CRLM undergoing resection. Baseline patient characteristics were presented in Supplementary Table 1. All readers completed both unaided and aided sessions, resulting in 3984 total assessments with no missing data. No protocol deviations occurred.

Clinician performance

The distribution of assessment responses and comparisons between the aided condition and the unaided condition were summarized in Table 1. For the primary endpoint, assistance with the prognostic tool significantly improved reader accuracy for predicting 3-year mortality (mean AUC difference, 0.091; 95% CI, 0.001 to 0.181; $P = 0.048$, two-sided; Table 2). Model assistance also improved the AUC across all secondary time-points for recurrence and mortality, with mean differences ranging from 0.091 to 0.116 (Table 2 and Fig. 1A). Importantly, the benefit of model assistance was

Table 2 | Comparison of model-based AUC, sensitivity, and specificity between the aided condition and the unaided condition

Model-based AUC improvement*		
Assessment categories	Mean, 95%CI	Nominal P
1-year recurrence risk	0.094 (8e−04, 0.188)	0.048
3-year recurrence risk	0.109 (−0.009, 0.227)	0.068
5-year recurrence risk	0.105 (−0.009, 0.220)	0.069
1-year mortality risk	0.116 (−0.006, 0.238)	0.060
3-year mortality risk	0.091 (0.001, 0.181)	0.048
5-year mortality risk	0.091 (−0.011, 0.193)	0.075
Model-based sensitivity improvement*		
Assessment categories	Mean, 95%CI	Nominal P
1-year recurrence risk	0.083 (−0.011, 0.176)	0.077
3-year recurrence risk	0.103 (−0.019, 0.226)	0.091
5-year recurrence risk	0.104 (−0.014, 0.223)	0.079
1-year mortality risk	0.105 (−0.016, 0.226)	0.083
3-year mortality risk	0.063 (−0.015, 0.142)	0.105
5-year mortality risk	0.089 (−0.014, 0.192)	0.083
Model-based specificity improvement*		
Assessment categories	Mean, 95%CI	Nominal P
1-year recurrence risk	0.023 (−0.011, 0.057)	0.178
3-year recurrence risk	0.079 (−0.052, 0.209)	0.218
5-year recurrence risk	0.008 (−0.022, 0.038)	0.607
1-year mortality risk	0.086 (−0.070, 0.242)	0.277
3-year mortality risk	0.068 (−0.014, 0.150)	0.095
5-year mortality risk	0.025 (−0.030, 0.079)	0.346

*Improvement (aided- unaided) was estimated using the standard MRMC method of Obuchowski and Rockette, where both the reader and the case are random effects.

consistent at the individual level, with every reader demonstrating improved diagnostic performance for the primary endpoint (Fig. 2) and others (Supplementary Figs. 1–5). The model’s standalone performance on the MRMC cohort was also reported for context (Supplementary Table 2).

Sensitivity was consistently improved with model assistance across most endpoints (mean differences range, 0.063 to 0.105), whereas specificity was not significantly changed (Table 2). Inter-reader agreement increased across all assessments (improvement range, 2.9% to 18.0%), with the largest gains observed for follow-up timing recommendation (18.0%) and 5-year mortality risk assessment (17.3%), indicating model assistance reduced variability among readers (Fig. 1B). Intra-reader consistency was lowest for 3 year-recurrence and mortality predictions, suggesting the model had its greatest influence on these reader’s assessments, while highest consistency on surgical support recommendations, suggesting minimal impact on reader’s decision making (Fig. 1C). Across physicians, using of the tool tended to improve the alignment between physician’s risk stratification and actual outcomes (Supplementary Figs. 6–11). In particular, model assistance enhanced their ability to correctly categorize high-risk patients into the ‘critical >70%’ group and low-risk patients into the ‘low <10%’ group.

Efficiency and decision confidence

Decision-making time was significantly reduced with model assistance (mean, 2.53 minutes vs. 3.04 minutes; nominal *P* < 0.001). Concurrently, reader confidence increased substantially, with the proportion of assessments rated “extremely high confidence” rising from 6.6% in the unaided condition to 52.5% in the aided condition (Fig. 1D). This increased confidence led to more polarized risk assessments, with physicians assigning more cases to ‘critical’ or ‘low’ risk categories. This translated into

recommendations for more frequent follow-up for high-risk patients but had minimal impact on decisions regarding surgical support (Supplementary Fig. 12).

Subgroup analyses

Across all experience levels, aided performance approached near-perfect accuracy (mean AUC > 0.98 for all assessments) (Fig. 3A). Junior and mid-level readers demonstrated the most substantial gains in AUC, whereas senior readers, who had high baseline accuracy, showed only modest improvement (Fig. 3B). A similar trend was observed for inter-reader agreement, which improved across all levels but most markedly among mid-level readers, who had the highest baseline variability (Fig. 3C). Senior physicians maintained highest overall consistency in assessments and decision-making (Fig. 3D). Importantly, the tool conferred universal workflow advantages, as all experience levels showed notable reductions in decision-making time and concurrent increases in confidence (Fig. 3E). The improvement in primary endpoint (AUC of 3-year mortality prediction) was consistent across all prespecified reader and case subgroup levels (Fig. 3F).

User experience survey

Post-study surveys confirmed positive user experience (100% response rate). Overall, 58.3% of readers were very confident about integrating the tool into their practice, 66.7% consider the tool moderately improved their efficiency of clinical decision-making, and 91.7% found it mostly reliable (Supplementary Table 3). Junior readers reported the greatest perceived gains in decision-making efficiency (75% reported significant improvement), and the highest overall satisfaction (Supplementary Table 3).

Discussion

We observed that the machine learning–based prognostic tool for CRLM¹⁴ was associated with improvements in clinician accuracy, and in exploratory analyses, with improvements in efficiency and decision confidence. This pilot randomized multi-reader multi-case study contributes to the evolving methodology for evaluating clinical prediction models within controlled but clinically contextualized trial settings. Rather than focusing solely on statistical model performance, our approach examines how AI assistance influences clinician performance^{20,21}. Within this structured experimental framework, these findings provide evidence of enhanced human–AI collaborative performance under controlled conditions. However, the observed improvements in efficiency and decision confidence were predefined as exploratory endpoints and should be interpreted as preliminary findings requiring confirmation in future prospective implementation studies to determine their impact on real-world surgical practice.

The primary finding of our study was the significant improvement in clinical accuracy for predicting 3-year mortality, a benefit that extended across secondary prognostic endpoints. Crucially, our analysis approach treated both cases and readers as random effects, ensuring the generalizability of findings to wider patient and physician groups—critical for informing real-world adoption^{22,23}. Our analysis revealed that this gain was largely driven by improved sensitivity, suggesting the tool is particularly effective at helping clinicians identify high-risk patients who might otherwise be overlooked—a crucial function for any prognostic aid. Specificity, while maintained, was not significantly improved, indicating the tool’s primary value lies in risk detection rather than in confirming low-risk status.

Furthermore, model assistance led to a marked increase in inter-reader agreement, suggesting that the tool can help standardize risk perception among clinicians, potentially reducing unwarranted practice variation and leading to more consistent patient management. In general, the CRLM prognosis prediction model reduced assessment time without compromising quality, while maintaining or improving confidence level. These improvements are especially impactful given the historical lack of validated, actionable prognostic tools for CRLM.

Our subgroup analyses revealed the tool’s greatest benefit emerged among less experienced clinicians, with junior surgical oncologists deriving

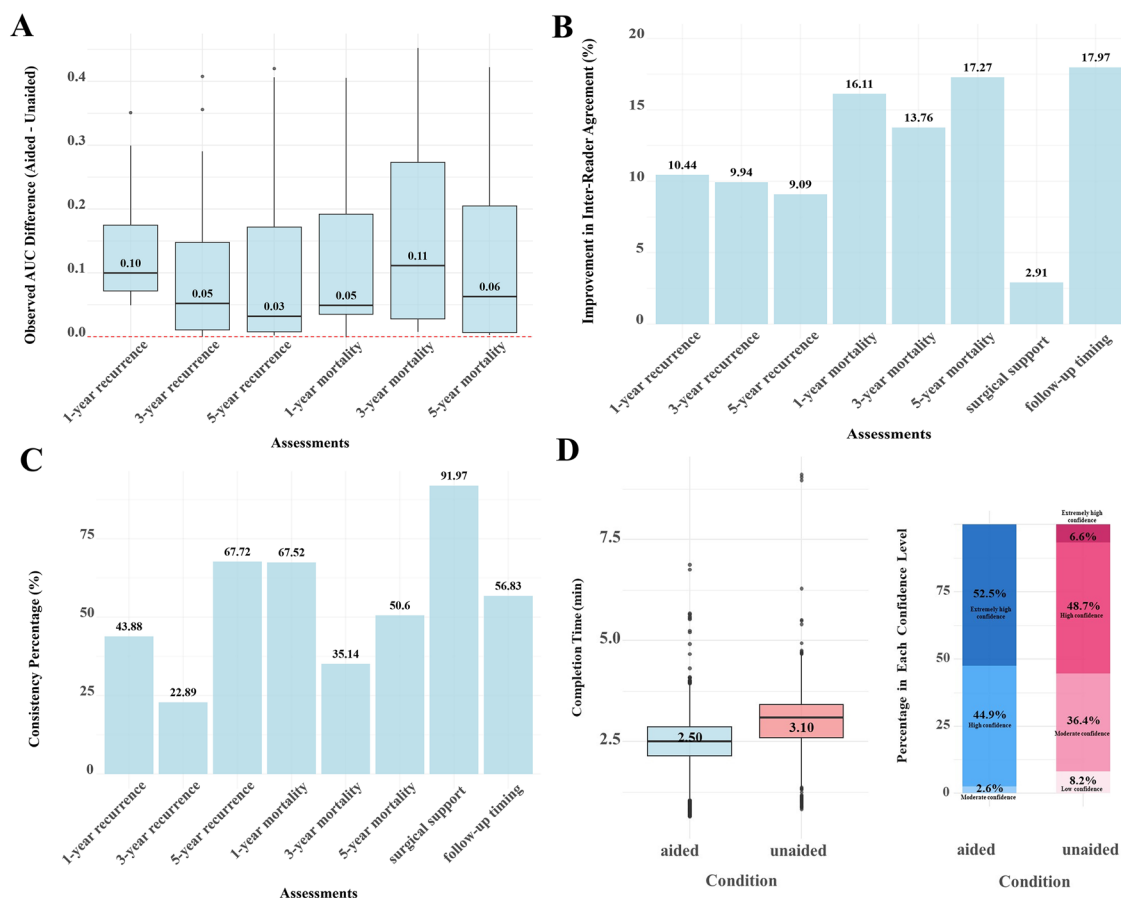


Fig. 1 | Overall Impact of Model Assistance on Clinician Performance.

A Distribution of the difference in observed AUC between aided and unaided conditions across all readers and assessments. **B** Percentage-point improvement in inter-reader agreement with model assistance by assessments: inter-reader agreement measures the consistency of assessments among different surgical oncologists for the same cases, indicating the reliability and standardization of clinical assessment and decision-making. **C** Distribution of intra-reader consistency across all

readers and assessments: intra-reader consistency measures how often individual surgical oncologists provide the same assessment for identical cases across different reading conditions, indicating the model's influence on individual decision-making. **D** Distribution of completion time and reader confidence level: completion time starts from retrieving case information till completing all the pre-specified assessment questions. Confidence level is the general confidence in risk assessments and decision-making of each patient.

the most pronounced gains in accuracy. This aligns with post-study feedback indicating that junior surgical oncologists were the most willing to continue using the tool. They reported that it helped structure their thinking and increased their confidence by providing a visual anchor for discussing prognosis. Conversely, senior and mid-career surgical oncologists, who already possess well-established internal heuristics, perceived the tool more as a confirmatory aid than a primary driver of decisions. In this context, the clinical prediction model can serve as a “digital mentor”, helping less experienced surgical oncologists approach the performance levels of their senior colleagues while maintaining efficiency gains for all users. Moreover, the tool appeared to promote consistency and reduce assessment variations in the mid-career surgical oncologists, a group that inherently exhibits greater variability in experience and practice patterns. The finding that decision-making time decreased significantly while confidence levels increased substantially addressed two critical barriers to clinical adoption: workflow disruption and decision uncertainty. These improvements were consistent across all experience levels, supporting the scalability of the approach.

This study highlights the importance of evaluating human–AI interaction when considering the translational impact of AI-powered prediction models. Although optimization of statistical metrics remains necessary, improvements in discrimination or calibration alone do not ensure meaningful influence on clinical behavior. Whether incremental gains in predictive performance translate into substantive changes in decision-making or patient outcomes remains an empirical question requiring prospective

real-world investigation. Our findings suggest that the downstream impact of AI assistance may be context-dependent. Model outputs may partially align with the tacit knowledge of experienced surgical oncologists, reinforcing rather than fundamentally altering established clinical judgment. In this setting, AI assistance may provide greater benefit to less experienced surgical oncologists by reducing variability and narrowing experience-related gaps. Moreover, real-world clinical decisions are often relatively insensitive to small differences in predicted risk. Modest changes in estimated survival probability may not meaningfully alter treatment strategies. More broadly, clinical decision-making is inherently multidimensional, requiring integration of quantitative predictions with comorbidities, functional status, psychosocial context, and patient preferences. Even highly accurate prediction models represent only one component of this complex process.

It is therefore important to distinguish improved clinician performance observed under controlled study conditions from definitive evidence of clinical utility. In this randomized MRC trial, we assessed the influence of AI assistance on prognostic assessment, decision consistency, efficiency, and confidence. While these findings provide structured evidence regarding human–AI collaborative performance, they do not directly establish improvements in patient outcomes, long-term practice patterns, or cost-effectiveness.

Several limitations should be acknowledged. First, the findings related to trust, reliability, and efficiency were derived from an exploratory questionnaire that has not undergone formal psychometric

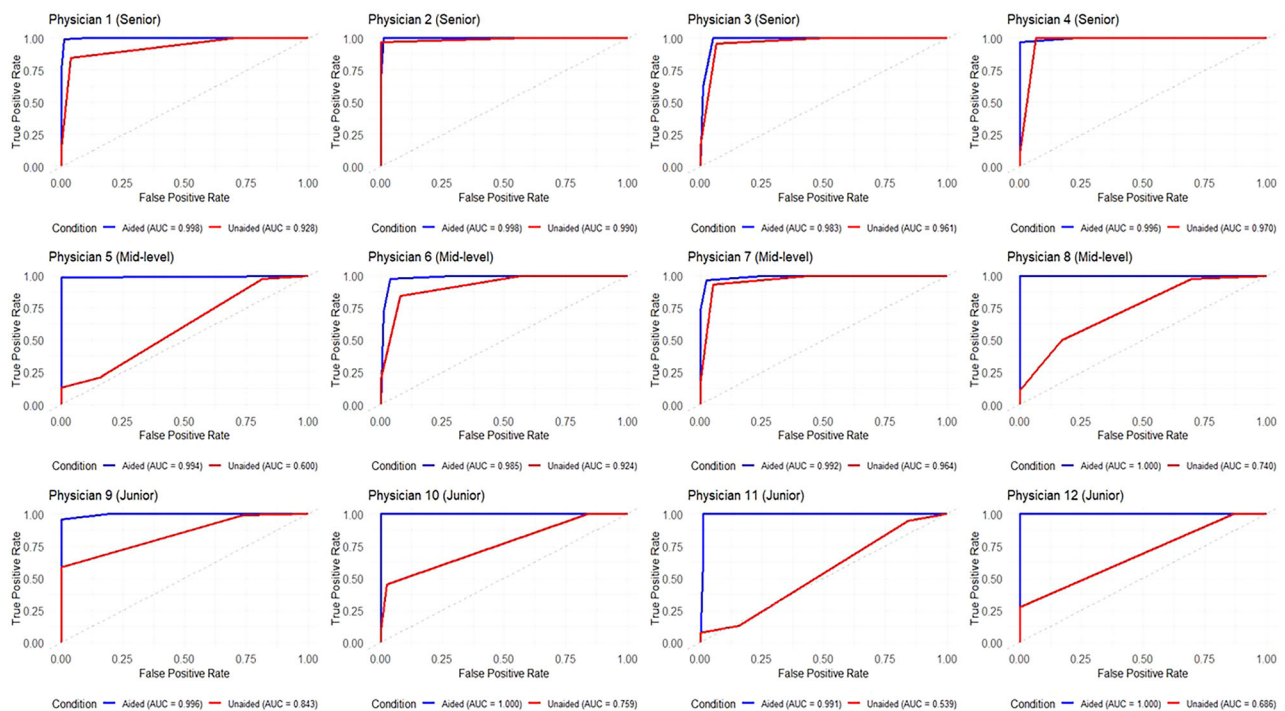


Fig. 2 | Individual Reader Performance for 3-year Mortality Prediction (4-point scale). Receiver operating characteristic (ROC) curves are shown for each of the 12 readers, stratified by experience level (Senior, Mid-level, Junior). Blue curves

represent performance with model assistance (aided); red curves represent performance without assistance (unaided).

validation. No validated instruments were used to assess these constructs. The absence of validated measurement tools may limit the robustness and generalizability of these results. Therefore, these findings should be interpreted with caution, and future studies should employ validated questionnaires to confirm and extend our observations. Second, as a single-center study, our findings may not be generalizable to institutions with different case mixes, practice patterns, or technological infrastructures. Multi-center studies are required to confirm these results in more diverse healthcare settings. Third, our analysis was based on retrospectively collected cases rather than prospective patient care, limiting our ability to assess the downstream consequences of physician decisions. Fourth, we were unable to evaluate sustained adoption or long-term workflow impact. Initial acceptance of new technologies may diminish without continued technical and organizational support^{24,25}.

Robust assessment of clinical utility requires evaluation beyond controlled MRMC studies. Future research should include prospective implementation designs—such as pragmatic cluster-randomized or stepped-wedge trials—comparing AI-assisted care with standard practice. Endpoints should extend beyond predictive accuracy to measurable changes in treatment decisions and clinically meaningful outcomes, including recurrence, survival, and quality of life, alongside workflow efficiency and cost-effectiveness. Multi-center external validation and prospective monitoring for performance drift will also be essential to ensure generalizability and long-term reliability. Together, these steps are necessary to establish true clinical utility and support responsible integration into routine practice.

In conclusion, by integrating user-centered design with prospective human–AI performance assessment in a randomized MRMC framework, we propose a structured approach for evaluating the impact of AI/ML prediction models on clinician performance. While this study does not establish definitive clinical utility, it represents an intermediate step between statistical model validation and real-world implementation studies. Bridging technical innovation with meaningful clinical benefit will require

longitudinal, prospective evaluation linking AI-assisted decisions to measurable patient outcomes.

Methods

Ethical approval and Study design

This was a pilot, single-center, prospectively designed and registered randomized multi-reader multi-case study conducted at the Cancer Hospital, Chinese Academy of Medical Sciences. The study employed a fully crossed design to evaluate how a validated machine learning prognostic tool for CLRM influences clinicians' performance. This study was conducted in accordance with the Declaration of Helsinki and was approved by the Institutional Review Board of the Cancer Hospital, Chinese Academy of Medical Sciences (IRB ID: NCC-017834). As this study involved no therapeutic interventions, it obtained an exemption for informed consent from the institutional review board. This study was registered at ClinicalTrials.gov (NCT07027605; Registration Date: January 1, 2025). The full protocol is provided in the Supplementary Material (Study Protocol).

The inclusion criteria of cases were defined as follows: patients (cases) aged ≥ 18 years who underwent colorectal resection with liver resection for histologically confirmed liver metastases from colorectal adenocarcinoma, and who were at least 5 years post-surgery at the initiation of the study. Exclusion criteria included a history of other malignancies or incomplete follow-up records. Cases were randomly selected from eligible patients in an independent dataset that was collected retrospectively from the Cancer Hospital, Chinese Academy of Medical Sciences, in the real world. Case selection was capped to achieve a balanced event and non-event cases for more efficient diagnostic evaluation. Readers were constituted of surgical oncologists from the Department of Surgical Oncology of the Digestive Tract with varying experience levels: junior (< 5 years), mid-level (5–10 years), and senior (> 10 years). Experience level was categorized by years of independent surgical practice (< 5 , 5–10, and > 10 years), consistent with the nationally recognized professional ranking system in Chinese tertiary hospitals. This stratification reflects progressive levels of clinical responsibility and seniority.

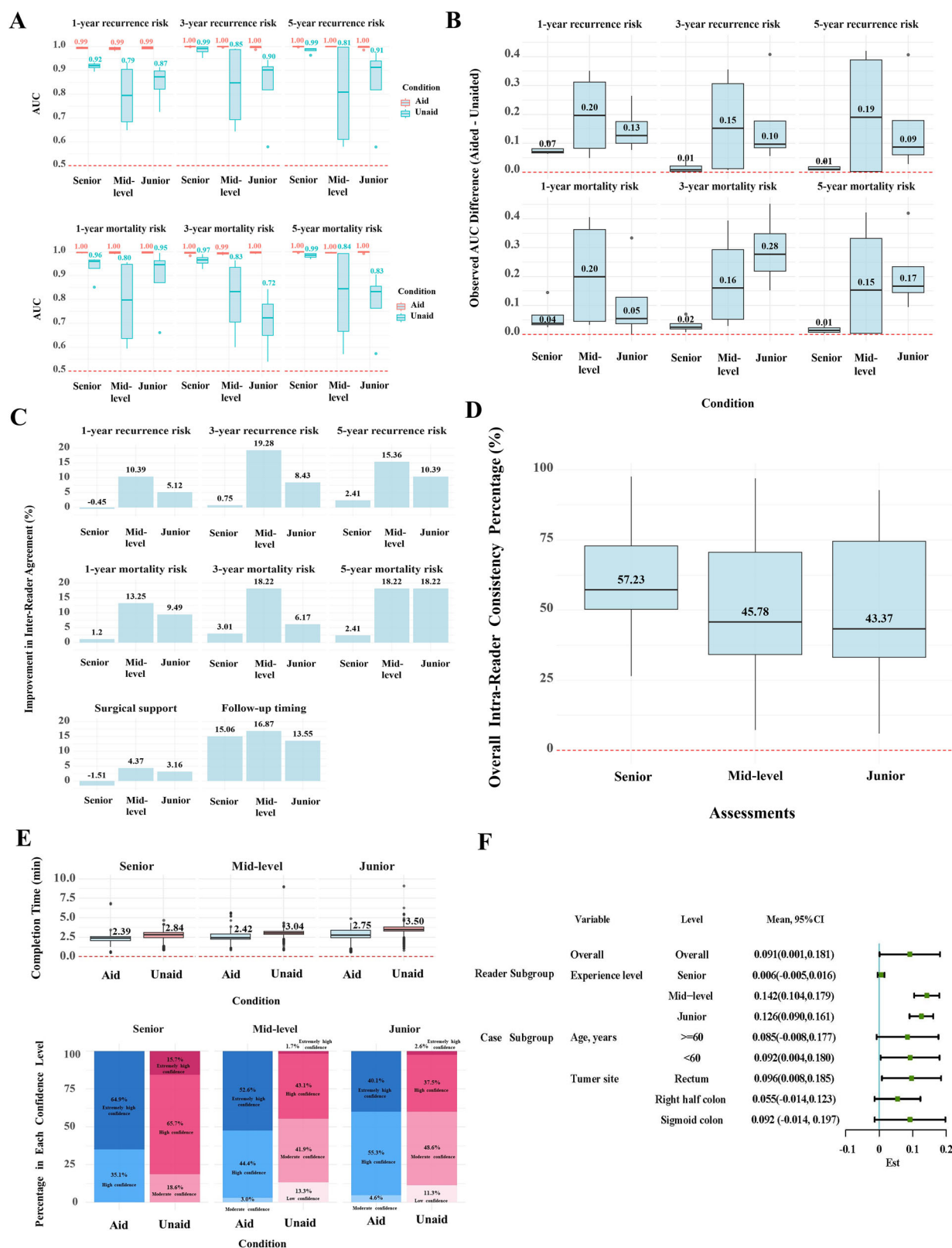


Fig. 3 | Impact of model assistance by reader experience. A Distributions of observed AUC in aided and unaided conditions. **B** Distributions of the difference in observed AUC between conditions. **C** Inter-reader agreement rate. **D** Overall intra-

reader consistency rate. **E** Completion time and confidence levels. **F** Forest plot of model-based AUC difference for the primary endpoint (3-years mortality prediction) across all prespecified subgroup levels.

Machine learning prediction tool

We utilized the models (<https://kanli.shinyapps.io/CRLMRP/>) developed in the previous paper¹⁴ to generate individualized risk predictions for post-operative recurrence and mortality. The model incorporated demographic,

clinical, laboratory, and genetic variables. Specifically, the prediction tool included 7 preoperative biochemical markers (fibrinogen, gamma-glutamyltransferase, red blood cell distribution width [RDW-SD and RDW-CV], neutrophil-to-lymphocyte ratio, AST/ALT ratio, and carcinoembryonic

antigen); 3 genetic markers (KRAS, BRAF, and mismatch repair status); and 23 general clinicopathological markers. The clinicopathological variables comprised age, sex, body mass index, comorbidities, American Society of Anesthesiology score, primary tumor site, tumor differentiation, lymph node metastasis, size of liver metastases, number of liver metastases, bilobar liver distribution, intraoperative radiofrequency ablation, preoperative chemotherapy, and postoperative chemotherapy, among others. The tool was previously validated in a multi-center study, demonstrating superior performance compared to traditional clinical risk scores¹⁴. Outputs include survival risk curves with landmark survival probabilities, presented through a user-friendly web interface.

Randomization and masking

Cases were randomized with a 1:1 ratio into Dataset A and Dataset B. Readers were stratified-randomized into Group A and Group B, preserving balanced experience levels. Randomization used computer-generated sequences to assign case order and reading conditions. The randomization applied to the allocation of cases (with and without AI assistance) to readers and to the reading sequence within a fully crossed design. Each reader evaluated all cases twice (unaided and aided with the prediction tool) during two sessions separated by a 5-week washout period to minimize recall bias. In Session 1, Group A evaluated Dataset A unaided and Dataset B aided, while Group B evaluated Dataset A aided and Dataset B unaided; assignments were reversed in Session 2 (see protocol for schema). Readers were blinded to patient outcomes (ground truths), identifiers, and peer evaluations.

Procedures

Each reader evaluated all cases twice during two sessions with and without the assistance of prediction tools, respectively. In the unaided condition, readers independently reviewed standardized perioperative electronic health records, including patient demographics, preoperative imaging findings, pathological characteristics of the primary tumor and liver metastases, laboratory results, and relevant surgical variables. Postoperative follow-up information, including recurrence and survival outcomes, was concealed. Based on the available clinical information, readers were instructed to estimate the individual patient's probability of recurrence and survival at predefined time points using their routine clinical reasoning process. No structured scoring system or additional guidance was provided, allowing clinicians to rely on their usual decision-making approach as applied in routine surgical practice. Predictions were recorded in a standardized electronic form to ensure consistency across readers and cases. In the aided condition, readers assessed the same information supplemented by real-time, tool-generated prognostic visualizations, including individualized survival curves. This entire process was managed through a purpose-built web application that seamlessly integrated patient data retrieval, model outputs, and digital evaluation forms into a single interface (Supplementary Material [Web-Based Assessment Interface]). The platform eliminated the need for additional software or manual data entry, thereby minimizing any additional workflow burden. Before the trial, all readers completed a standardized training session on case data extraction, interpretation of the model prediction outputs, navigation of the web application, and use of the digital evaluation forms. The readers were instructed to answer the questions and provide their own assessment solely based on their independent review and synthesis of the raw electronic health records review and the model-predicted information described above.

For each case, readers assessed: risk predictions for 1-, 3-, and 5-year recurrence and mortality risk (4-point scale: low <10%, moderate 10%–30%, high 30%–70%, critical >70%; the risk bins were established by expert consensus to reflect distinct clinical management pathways), recommendations for surgical procedures and follow-up intervals; decision confidence (protocol-defined scale); and electronically recorded evaluation time (Supplementary Material [Patient Evaluation Questionnaire]).

Outcomes

The primary endpoint is the difference in area under the receiver operating characteristic curve (AUC-ROC) for 3-year mortality prediction between aided and unaided conditions. The secondary endpoints include differences in AUC-ROC for 1-year and 5-year mortality, and 1-, 3-, and 5-year recurrence; differences in sensitivity and specificity for all time points; and inter-rater reliability for prognostic assessments. Exploratory endpoints include inter-rater reliability for treatment and follow-up recommendations, differences in decision confidence, and time spent between conditions.

Trial oversight

The study was overseen by the institution with protocol adherence monitoring. No data monitoring committee was required due to the retrospective case evaluation and absence of anticipated safety concerns for readers and cases. All case data were fully anonymized to ensure patient confidentiality. An unblinded statistician conducted the final analysis.

Statistical analysis

Efficacy was assessed in all randomized readers and cases. No safety analysis was planned in the study. Time-dependent AUC-ROCs for predictions were estimated using the dynamic ROC curve approach with inverse probability of censoring weighting for right-censored data²⁶. Between-condition differences were analyzed using the Obuchowski-Rockette (OR) method²⁷, treating both reader and case as random effects for generalizability. Sensitivity and specificity differences were assessed via the OR method. Inter-rater reliability, decision confidence, and time spent were summarized descriptively. The planned sample size (12 readers and 166 cases) provided > 80% power to detect a 0.05 absolute AUC difference for 3-year mortality prediction at $\alpha = 0.05$ (two-sided). Hypothesis testing was limited to the primary endpoint without multiplicity adjustments; nominal *P* values were reported for secondary analyses. Subgroup consistency (e.g., by reader experience or case characteristics) was assessed using mean AUC differences and 95% confidence intervals via the OR method. Missing data was managed via protocol-specified data handling and analysis guidelines. Analyses used R software (version 4.3). The full statistical analysis plan is in Section 6 of the protocol.

Post-study Surveys

After completing the study, all readers were administered a 9-item post-study survey to assess their user experience (Supplementary Material [Clinical Tool Usage and Evaluation Survey]). The structured questionnaire evaluated the tool's perceived utility, usability, reliability, and workflow integration. Responses were aggregated to identify key trends in clinician satisfaction and barriers to adoption.

Reporting Guidelines

This study was reported in accordance with the CONSORT-AI 2020 extension²⁸ for clinical trials involving artificial intelligence interventions. A completed CONSORT-AI checklist is provided in the Supplementary Material.

Data availability

The datasets generated and/or analyzed during the current study are not publicly available due to restrictions designed to protect patient privacy and in accordance with the ethical approval governing this study, but are available from the corresponding author on reasonable request.

Code availability

The code generated in this study will be made available by the corresponding author upon reasonable request.

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Author contributions

Concept and design: Acquisition, analysis, or interpretation of data: Qichen Chen, Jinliang Tong, Yiqiao Deng, Xinyu Bi, Yuan Li, Kan Li, and Hong Zhao Drafting of the manuscript: Qichen Chen, Jinliang Tong, Yiqiao Deng, Yuan Li, and Kan Li Critical review of the manuscript for important intellectual content: Qichen Chen, Yuan Li, Kan Li, and Hong Zhao Statistical analysis: Qichen Chen, Jinliang Tong, Yiqiao Deng, and Kan Li Administrative, technical, or material support: All authors Supervision: Qichen Chen, Kan Li, Hong Zhao and Yuan Li.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Qichen Chen, Yuan Li, Kan Li or Hong Zhao.

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